Team Involvement in Demonstration

Date	21 October 2026
Team ID	TDS-AI-2026
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	3 Marks

Project Title

Personalized Travel Itinerary Planner

Team Composition

Role	Responsibility
Full-Stack Developer	Complete development of backend API, frontend UI, database schema, AI integration, and deployment
Project Lead	Architecture decisions, technology selection, feature prioritization, documentation

Development Contribution Breakdown

Backend Development

- Designed and implemented the FastAPI application structure (main.py, routers, services, repositories)
- Implemented JWT authentication using Supabase JWKS with key caching
- Built the AI service with Gemini API integration, Groq fallback, and retry logic
- Developed Pydantic schemas with custom validators for all data models
- Created the generation service with atomic transaction handling (rollback on failure)
- Wrote the database migration script (001_init.sql) with RLS policies and helper functions
- Implemented request logging middleware with unique request IDs

Frontend Development

- Set up React 19 + TypeScript + Vite project with TailwindCSS v4
- Implemented Supabase Auth integration (login, signup, session management)
- Built all 5 application pages (Login, Signup, Dashboard, New Trip, Trip Detail)
- Integrated React Query for server state management with optimistic updates
- Implemented drag-and-drop activity reordering using dnd-kit
- Created the Activity Edit Dialog with form validation
- Designed the travel-themed login/signup UI with SVG backgrounds

Database & Security

- Designed the 6-table PostgreSQL schema optimized for the use case
- Implemented 14 Row-Level Security policies for complete data isolation
- Created 4 SECURITY DEFINER helper functions for safe policy evaluation
- Set up database triggers for auto-profile creation and timestamp updates

AI & Prompt Engineering

- Designed the itinerary generation prompt with structured output enforcement
- Created the single-day regeneration prompt with duplicate avoidance
- Tuned AI parameters (temperature 0.7, max 8192 tokens) for best results
- Implemented robust output validation pipeline

Testing

- Wrote 22 unit tests covering schema validation and service layer logic
- Tested all API endpoints end-to-end with deployed services

Deployment & DevOps

- Deployed frontend to Vercel with proper SPA routing configuration
- Deployed backend to Render with environment variable management
- Configured CORS for production origins
- Set up Supabase project with database and authentication

Demonstration Roles

Segment	Presenter	Duration
Project Introduction & Problem Statement	Team Lead	2 min
Technology Stack Overview	Developer	2 min
Live Demo — Authentication & Dashboard	Developer	3 min
Live Demo — AI Itinerary Generation	Developer	5 min
Live Demo — Customization Features	Developer	5 min
API Documentation (Swagger UI)	Developer	2 min
Architecture & Security Explanation	Team Lead	3 min
Scalability & Future Plans	Team Lead	2 min
Q&A	Both	5 min

Preparation Checklist

- [x] Live application deployed and accessible on Vercel
- [x] Backend API deployed and accessible on Render
- [x] Swagger API docs accessible at /docs
- [x] Test account created with sample trips pre-generated
- [x] All environment variables configured in production
- [x] End-to-end tested with deployed services
- [x] Presentation slides prepared
- [x] Demo script rehearsed
- [x] Backup plan: screenshots of all features in case of network issues